

Adam Mrotek

Full-Stack Engineer

adam.mrotek@gmail.com | +44 7517 624501 | linkedin.com/in/adam-mrotek | github.com/AdamMrotek | adammmrotek.github.io/portfolio | villageos.co.uk

SUMMARY

Full-stack engineer with ~4 years' commercial experience building complex, user-facing web applications end-to-end. Proven track record of owning delivery from product spec to production and working closely with design to ship well-architected, type-safe software. Strong in TypeScript, React and Next.js with a modern front-end toolchain (TanStack Query, shadcn/ui) and hands-on backend in Python / FastAPI, and an active focus on building real products on top of LLMs.

TECHNICAL SKILLS

TypeScript, React, Next.js, TanStack Query, shadcn/ui, Node.js, Python, FastAPI, PostgreSQL, Tailwind CSS, AWS, Docker

WORK EXPERIENCE

Founder & Full-Stack Engineer, VillageOS (Self-employed)

Apr 2026 - Present

Independent product engineering · live in production (public demo + deployed API) · villageos.co.uk

Next.js, React, TypeScript, TanStack Query, shadcn/ui, FastAPI, AWS Lambda, Supabase, OpenAI, Groq, PostHog

Designing, building and operating VillageOS solo, a live AI product that turns scattered parent communication - WhatsApp threads, PDF newsletters, photos of flyers - into schema-valid calendar events with action items. Owned every layer end-to-end: frontend, backend, data, infrastructure, CI/CD and LLM evaluations.

- **Typed on both sides** - Pydantic v2 is the single source of truth for the LLM extraction boundary (structured output) and FastAPI request/response validation; the TypeScript client mirrors the same shapes, so the contract is enforced at the API edge.
- **Structured AI, not vibes** - instructor + Pydantic force a strict schema on every extraction, across multimodal (text + vision) inputs, with a golden-dataset eval harness tracking per-field accuracy, latency and cost.
- **Real infrastructure** - FastAPI on AWS Lambda (arm64) via SAM, Supabase JWT auth tied to Postgres Row Level Security, deployed through OIDC GitHub Actions.
- **Online A/B in production** - a server-assigned PostHog experiment on the live extraction model, field-edit-rate metric with latency and cost guardrails - closing the loop from offline eval to real user behaviour.

Full-Stack Engineer, Croud

Jul 2023 - Mar 2026 (2y 9m)

Marketing Agency

React, TypeScript, Next.js, TanStack Query, shadcn/ui, FastAPI, Tailwind CSS, Vue 3, Docker, Claude / MCP

- **Core Platform (Croud Control, 3,000+ users):** Owned front-end delivery across a marketing-automation suite serving 3,000+ users; co-architected the TypeScript component design system (React + Tailwind + shadcn/ui + Storybook) and modernised legacy surfaces to type-safe patterns, accelerating feature delivery across product areas.
- **Social Influencer Platform:** Technical owner of key React (TanStack Query + shadcn/ui) front-end features and their supporting FastAPI backend for an in-house influencer-marketing platform - an internal build that replaced a £17M third-party agency quote - delivering user-facing surfaces and API data modelling against strict Figma specs.
- **Taxonomy Tool:** Built an internal tag-alignment app (front-end plus supporting backend) projected to save ~£100k/month by removing cross-team tagging inconsistency and cleaning data for downstream analytics.
- **AI Tooling:** Built the front-end apps that showcased internal AI tools and agentic pipelines in client pitches, and shipped a custom Claude sub-agent that converts Figma designs directly into our codebase - driving cross-team adoption and supporting new-business upsell.

Customer Engineer, Hullabalook

Apr 2022 - Jul 2023 (1y 4m)

Improving retail experience using data analytics · Retail · B2B · SaaS · eCommerce

React, Vue.js, Vuex, Nuxt, TypeScript, Fabric.js, Python, GCP

- Drove successful client upsells by prototyping new features, and improved Lighthouse / SEO scores to lift app performance across multiple clients.
- Shipped features and multiple apps within cross-functional product teams, delivering through CI/CD pipelines.
- Built robust applications using React, Vue, Vuex and Nuxt, plus highly visual interfaces built with Fabric.js and the HTML5 Canvas API.
- Improved data accuracy by building programmatic solutions for unstructured, incomplete datasets using Python, Google Colab and GCP.
- Mentored junior engineers through code reviews and pairing sessions, helping raise code quality and shorten onboarding.

Architectural Assistant (Part I & II), Bryden Wood

Mar 2018 - Sep 2021 (3y 6m)

Multidisciplinary architectural company · Property · Design · Construction

- Designed and contributed to architectural projects for airports and data centers, applying platform-based, systems-thinking design principles.
- Facilitated cross-functional collaboration between architectural and mechanical / electrical design teams to achieve project objectives.

EDUCATION

Anthropic Academy

May 2026

Certificate of completion - Claude with Amazon Bedrock

- Building production-ready AI applications with Claude on Amazon Bedrock - the Bedrock API, prompt engineering and evaluations, tool use, retrieval-augmented generation (RAG), the Model Context Protocol (MCP), and agents.

MSc & BArch, Architecture

2011 - 2017

Politechnika Poznańska (Poznan University of Technology) · incl. Erasmus, Università di Firenze

ADDITIONAL

Languages: English (Full Professional), Polish (Native / Bilingual)

Currently learning: AI pipelines and multi-agent workflows, RAG, and app deployment.

Interests: Parenthood tricks, feng-shui, cooking.